# Project Report

## Title: Topic: Designing an Interconnected Multi-Building Network

***Course Details***
**Course Title:** Computer Networks
**Couse Code**: CSE 405
**Section:** 05

***Submitted To***
Md. Khalid Mahbub Khan
Lecturer
Department of Computer Science and Engineering

***Submitted By***
Farhatun Nahar Priya
Student ID: 2023-1-60-202

*Date of Submission: 15th December 2025.*

# Table Of Content

1. Problem Statement..... 2
2. Network Design ..... 3
2.1 Design Overview ..... 3
2.2 IP Addressing Scheme ..... 4
2.4 DHCP Configuration and Automatic PC Addressing..... 13
2.6 DNS Server Configuration..... 24
2.7 Web Server Configuration..... 25
3. Experimental Results ..... 26
3.1 Inter-Facility Connectivity Test (PC to PC Ping Verification) ..... 26
3.2 Web Access Verification ..... 28
4. Conclusion ..... 33

# 1. Problem Statement

Modern educational institutions require reliable, secure, and scalable computer networks to support academic, administrative, and research activities. As the number of users and campus facilities increases, isolated networks become inefficient, difficult to manage, and prone to configuration errors. Therefore, a well structured interconnected campus network is essential to ensure seamless communication, centralized services, and efficient resource sharing.

This project focuses on the design and implementation of an interconnected multi-building network for a university campus consisting of five key facilities: Main Campus, Library Center, Computer Lab, Faculty Offices, and Administration Block. Each facility is required to operate as an independent network segment with its own router while maintaining full inter-facility communication.

The network is designed using Class A IP addressing for LAN segments and Class B IP addressing for inter-router communication. Dynamic routing is implemented using EIGRP, automatic IP address allocation is ensured through DHCP servers, and centralized DNS and Web services are hosted in the Administration Block. Finally, successful network operation is verified through practical testing, including ping operations and web access validation.

# 2. Network Design

## 2.1 Design Overview

The proposed network follows a hierarchical and modular campus network design. Each facility operates as an independent Local Area Network (LAN) connected through a dedicated router. All facility routers are interconnected via the Administration Block Router, which acts as the core router of the entire network.

This centralized core based design improves scalability, simplifies troubleshooting, and ensures efficient routing among all facilities. Centralized services such as DNS and Web Server are also hosted in the Administration Block, enabling campus wide accessibility.

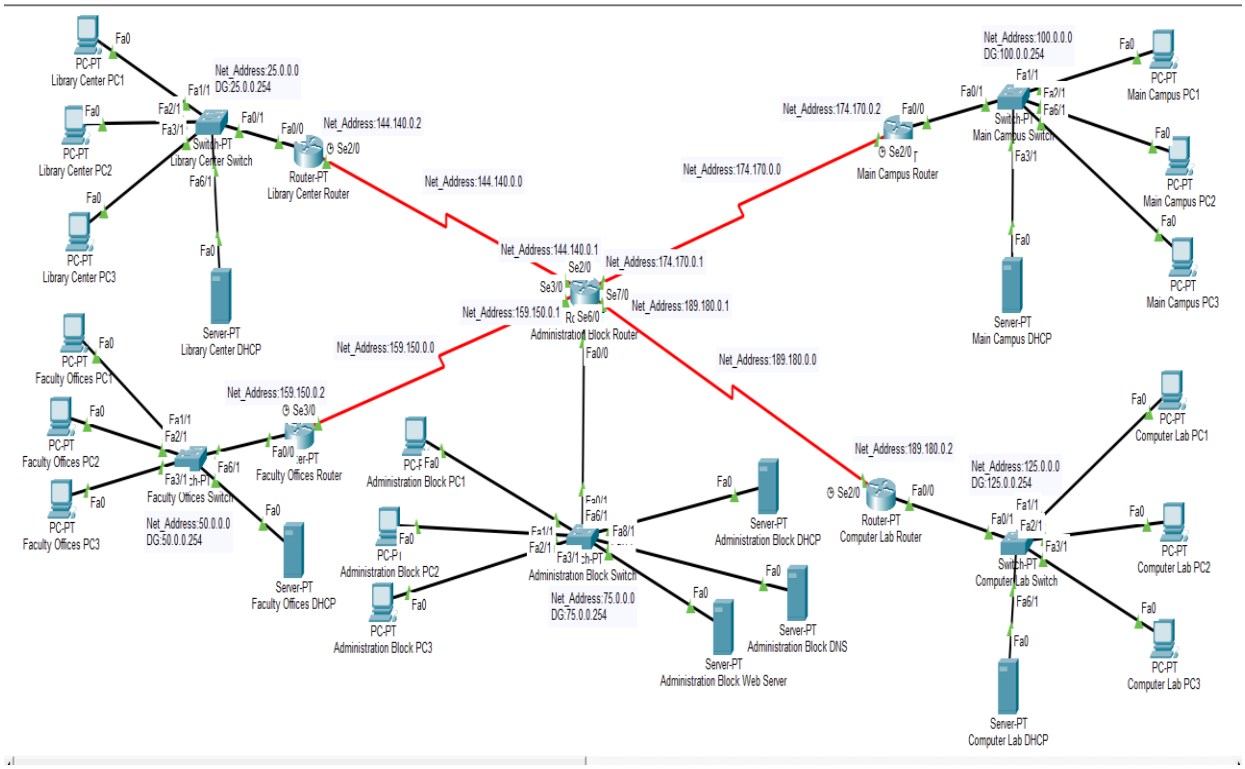



**Figure 1: Overall Interconnected Campus Network Topology**

## 2.2 IP Addressing Scheme

To ensure efficient address management and compliance with the project requirements, Class A IP addressing is used for all LAN segments, while Class B IP addressing is used for inter-router serial links. Class C addressing is intentionally avoided. Proper subnetting is applied to accommodate host requirements and ensure scalability.

| Facility | Network Address | Subnet Mask | Default Gateway | IP Class |
|---|---|---|---|---|
| **Main Campus** | 100.0.0.0 | 255.0.0.0 | 100.0.0.254 | Class A |
| **Library Center** | 25.0.0.0 | 255.0.0.0 | 25.0.0.254 | Class A |
| **Faculty Offices** | 50.0.0.0 | 255.0.0.0 | 50.0.0.254 | Class A |
| **Computer Lab** | 125.0.0.0 | 255.0.0.0 | 125.0.0.254 | Class A |
| **Administration Block** | 75.0.0.0 | 255.0.0.0 | 75.0.0.254 | Class A |

**Table 1: LAN IP Addressing**

| Link | Network Address | Subnet Mask | IP Class |
|---|---|---|---|
| Library to Administration Block | 144.140.0.0 | 255.255.0.0 | Class B |

| | | | |
|---|---|---|---|
| Faculty Offices to Administration Block | 159.150.0.0 | 255.255.0.0 | Class B |
| Main Campus to Administration Block | 174.170.0.0 | 255.255.0.0 | Class B |
| Computer Lab to Administration Block | 189.180.0.0 | 255.255.0.0 | Class B |

**Table 2: Inter Router Serial Link (Serial DCE) Addressing**

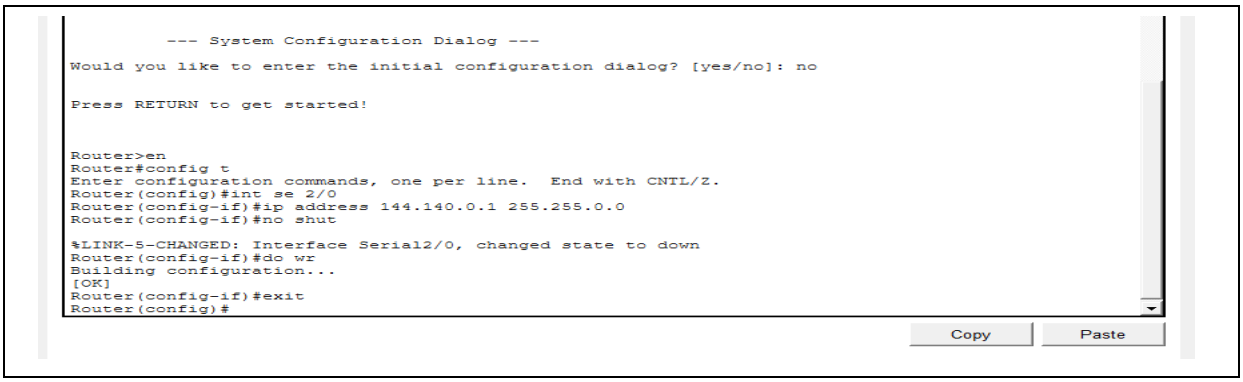
## 2.3 Router Interface Configuration

The first step of the network implementation involved configuring all router interfaces with static IP addresses. FastEthernet interfaces were configured to serve as default gateways for their respective LANs, while Serial interfaces were assigned Class B IP addresses to enable inter-router communication. Serial DCE interfaces were configured with appropriate clock rates to ensure proper synchronization. This initial configuration established the foundational connectivity required for higher-level services.

### ➢ Router Configurations for Administration Block

### Se2/0

Router>en
Router#config t
Router(config)#int se 2/0
Router(config-if)#ip address 144.140.0.1 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit



## Se 3/0

Router(config)#int se 3/0
Router(config-if)#ip address 159.150.0.1 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int se 3/0
Router(config-if)#ip address 159.150.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Copy Paste

## Se6/0

Router(config)#int se 6/0
Router(config-if)#ip address 189.180.0.1 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int se 6/0
Router(config-if)#ip address 189.180.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial6/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Copy Paste

## Se7/0

Router(config)#int se7/0
Router(config-if)#ip address 174.170.0.1 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int se7/0
Router(config-if)#ip address 174.170.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial7/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Copy Paste

## Fa0/0

Router(config)#int fa0/0
Router(config-if)#ip address 75.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int fa0/0
Router(config-if)#ip address 75.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#
```

Copy Paste

Administration Block Router

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 144.140.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se 3/0
Router(config-if)#ip address 159.150.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se 6/0
Router(config-if)#ip address 189.180.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial6/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se7/0
Router(config-if)#ip address 174.170.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial7/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 75.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

Copy Paste

Top

**Figure 2: Router Interface Configuration (CLI) for Administration Block**

## ➢ Router Configurations for Library Center

## Se2/0

Router>en
Router#config t
Router(config)#int se 2/0
Router(config-if)#ip address 144.140.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config)#exit

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 144.140.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
```

## Fa0/0

Router#en
Router#config t
Router(config)#int fa0/0
Router(config-if)#ip address 25.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#
Router(config-if)#exit

```
Router(config)#int fa0/0
Router(config-if)#ip address 25.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
```

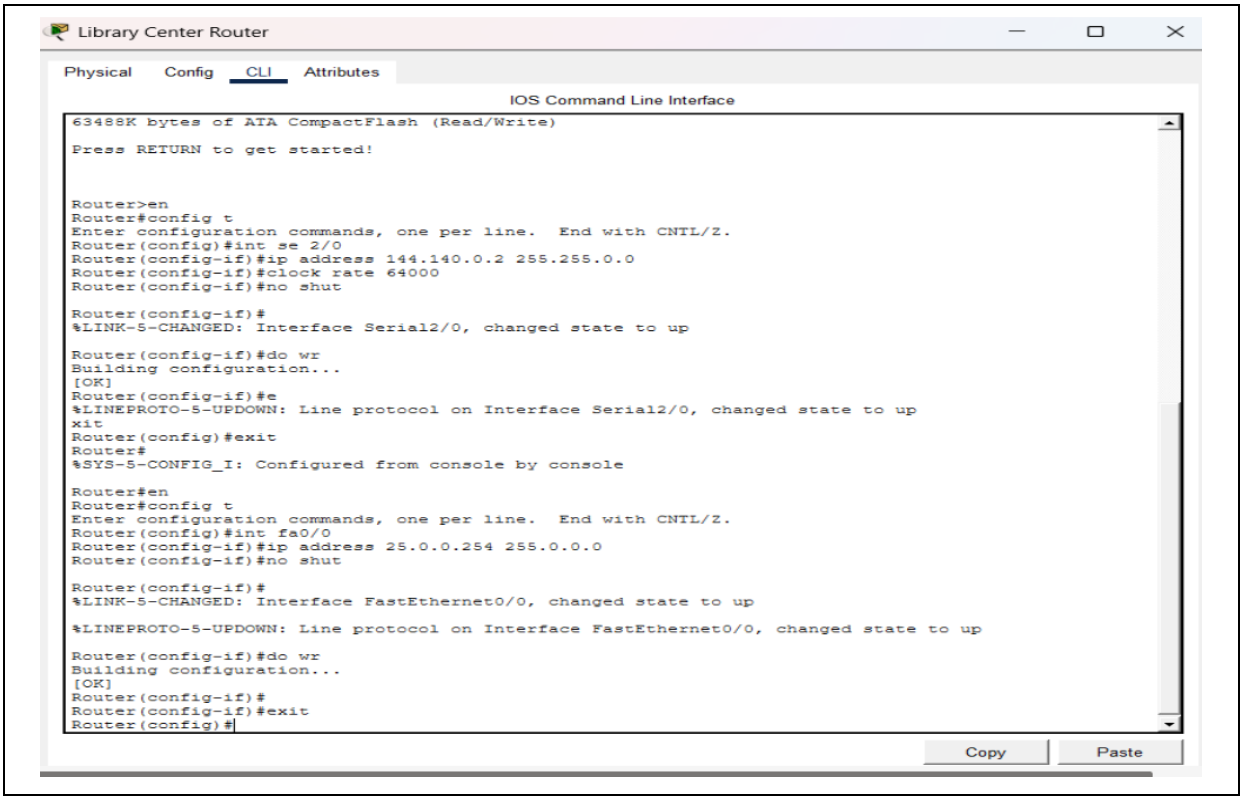
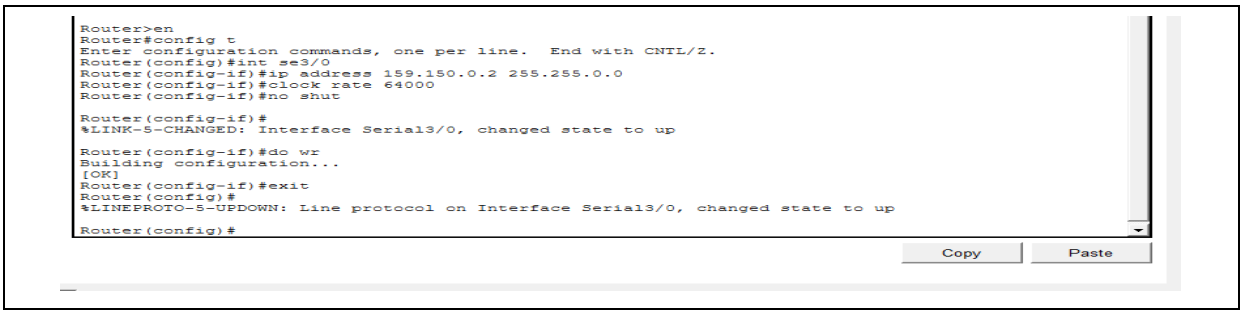



**Figure 3: Router Interface Configuration (CLI) for Library Center**

## ➢ Router Configurations for Faculty Offices

### Se3/0

Router>en
Router#config t
Router(config)#int se3/0
Router(config-if)#ip address 159.150.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit



## Fa0/0

Router(config)#int fa0/0
Router(config-if)#ip address 50.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int fa0/0
Router(config-if)#ip address 50.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Copy Paste

Faculty Offices Router

Physical Config CLI Attributes

IOS Command Line Interface

```
Processor board ID PT0123 (0123)
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)

Press RETURN to get started!


Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se3/0
Router(config-if)#ip address 159.150.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config)#int fa0/0
Router(config-if)#ip address 50.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Copy Paste

**Figure 4: Router Interface Configuration (CLI) for Faculty Offices**

## ➢ Router Configurations for Main Campus

## Se2/0

Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 174.170.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 174.170.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#
```

Copy Paste

## Fa0/0

Router(config)#int fa0/0
Router(config-if)#ip address 100.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int fa0/0
Router(config-if)#ip address 100.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Copy Paste

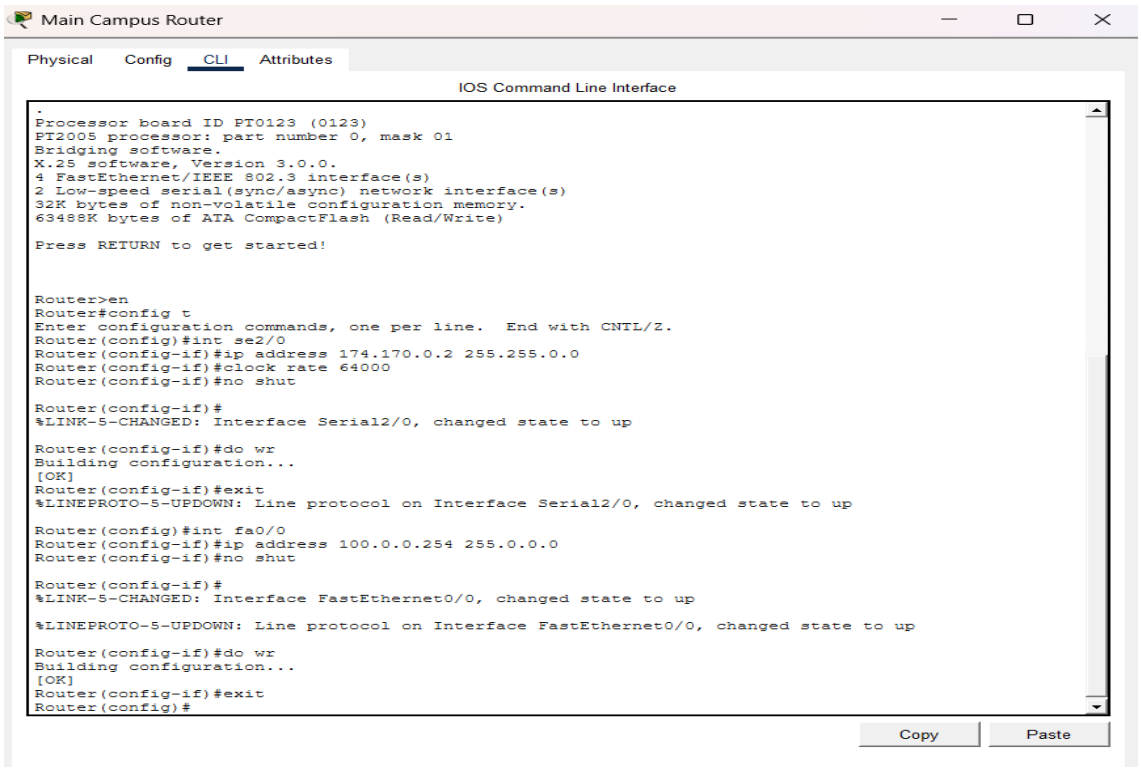



**Figure 5: Router Interface Configuration (CLI) for Main Campus**

## ➢ Router Configurations for Computer Lab

### Se2/0

Router>en
Router#config t
Router(config)#int se 2/0
Router(config-if)#ip address 189.180.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 189.180.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#
```

Copy Paste

## Fa0/0

Router(config)# int fa0/0
Router(config-if)#ip address 125.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)# int fa0/0
Router(config-if)#ip address 125.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Copy Paste

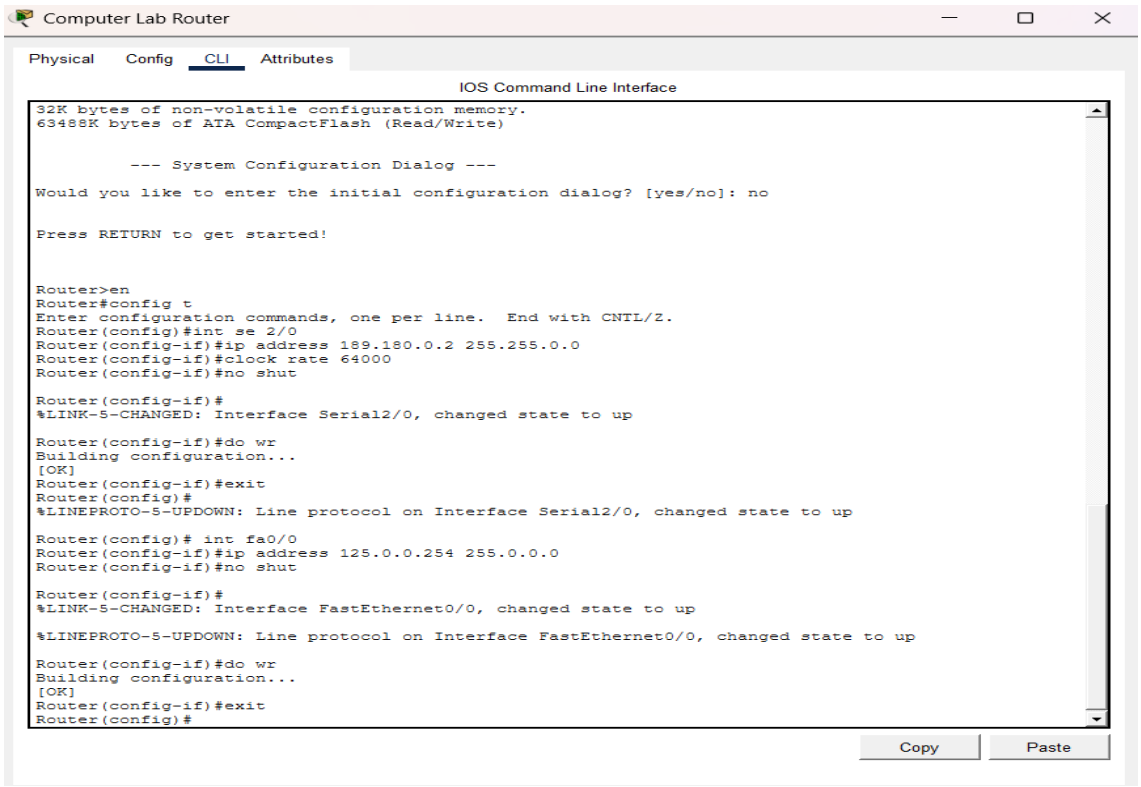



**Figure 6: Router Interface Configuration (CLI) for Computer Lab**

## 2.4 DHCP Configuration and Automatic PC Addressing

After completing router interface configuration, DHCP services were configured in each facility. Dedicated DHCP servers were deployed to automatically assign IP addresses, subnet masks, default gateways and DNS server information to client PCs.

As a result, all PCs were configured automatically without manual IP assignment, ensuring consistency and minimizing configuration errors.

### ➢ Administration Block DHCP

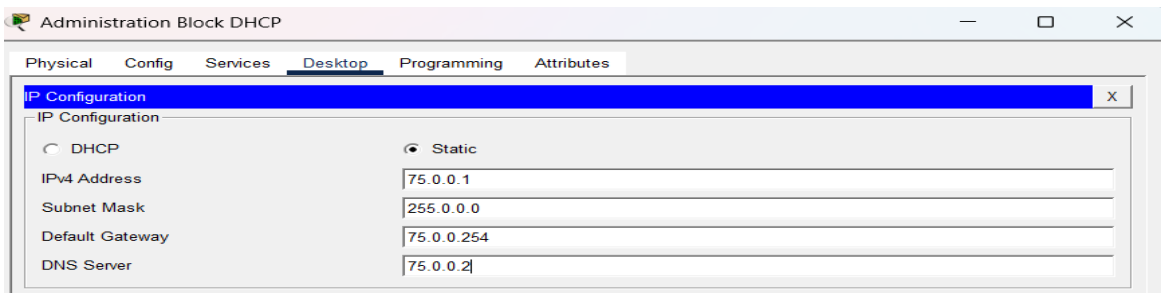



**Figure 7: DHCP Configuration for Administration Block**

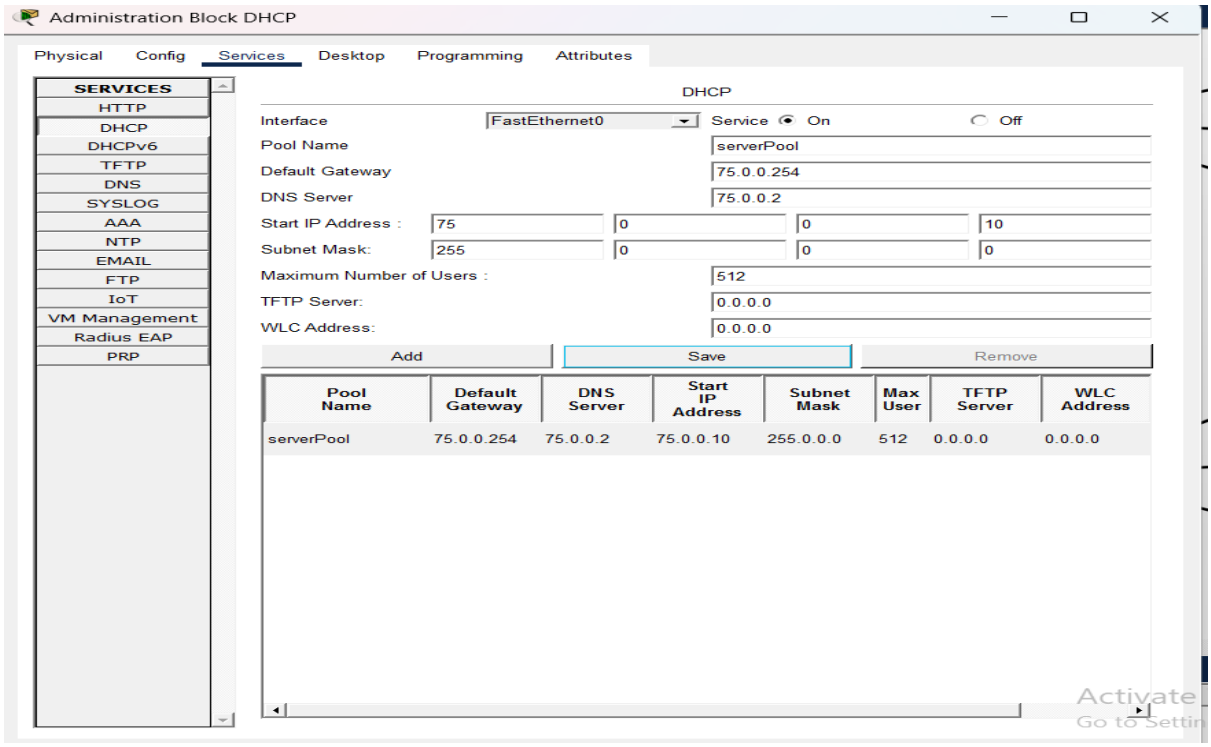



**Figure 8: DHCP Service Running for Administration Block**

## ➢ Automatic IP Assignment to PCs in Administration Block

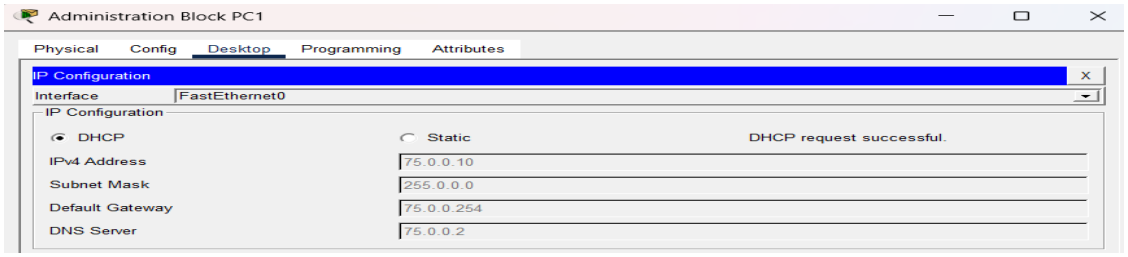



**Figure 8.1: Automatic DHCP IP Assignment to PC1 in Administration Block**

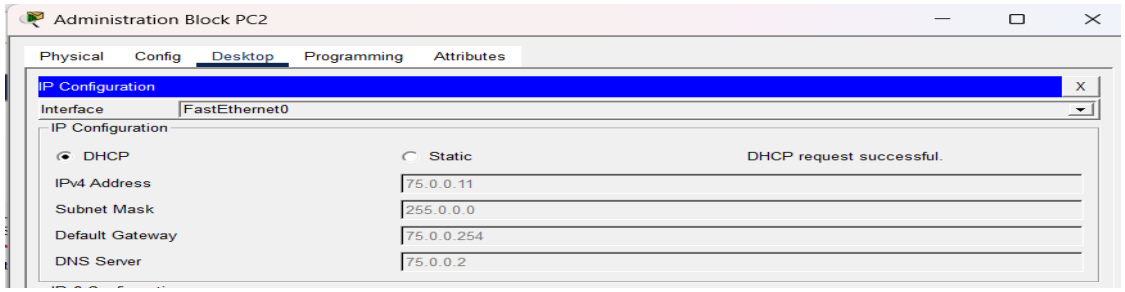



**Figure 8.2: Automatic DHCP IP Assignment to PC2 in Administration Block**

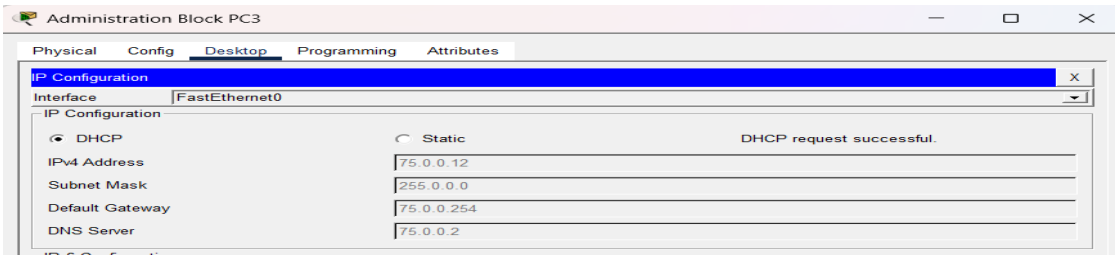



**Figure 8.3: Automatic DHCP IP Assignment to PC3 in Administration Block**

## ➢ Library Center DHCP

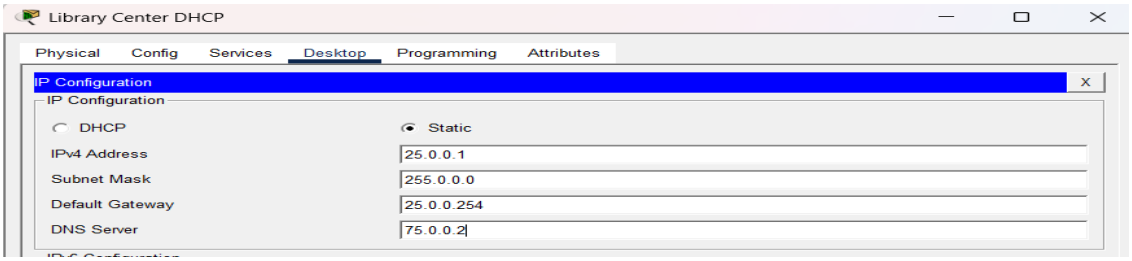



**Figure 9: DHCP Configuration for Library Center**

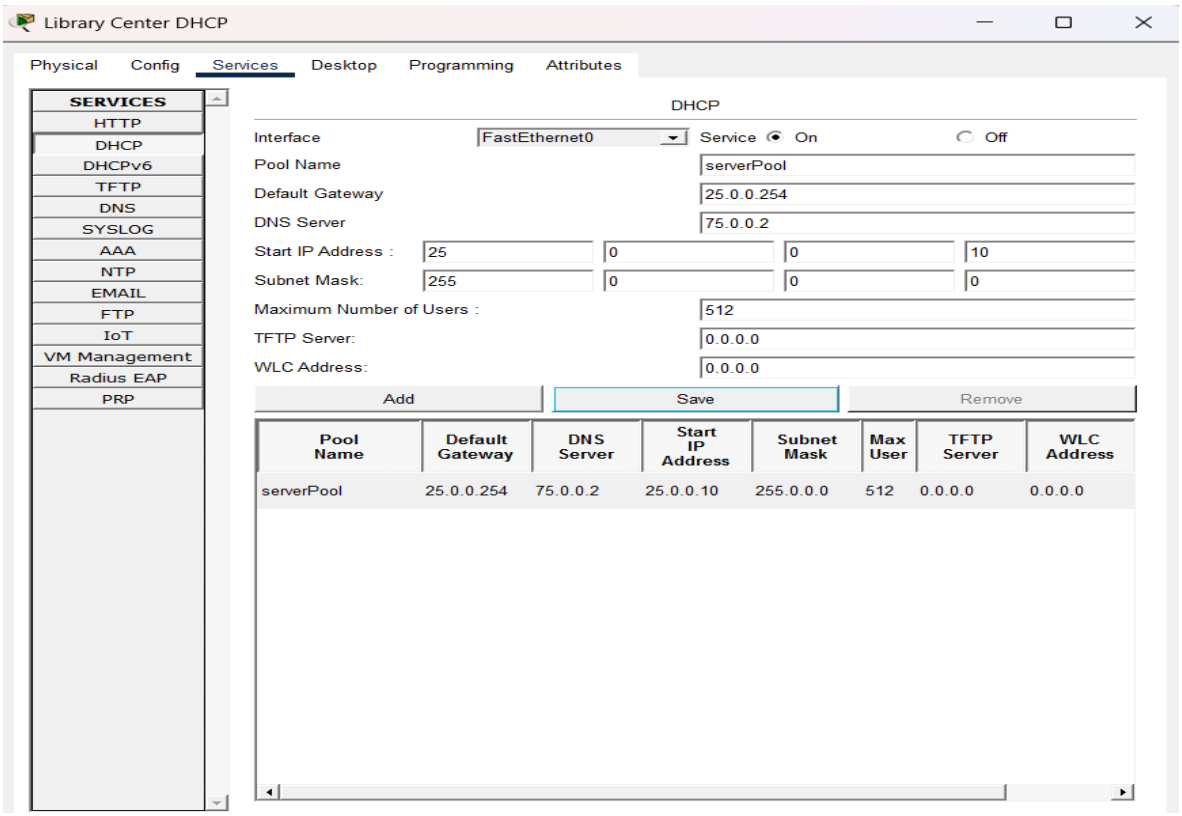



**Figure 10: DHCP Service Running for Library Center**

## ➢ Automatic IP Assignment to PCs in Library Center

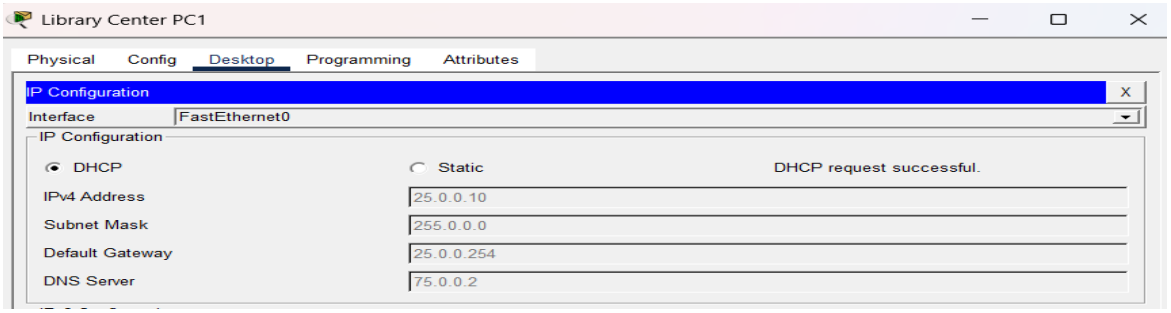



**Figure 10.1: Automatic DHCP IP Assignment to PC1 in Library Center**

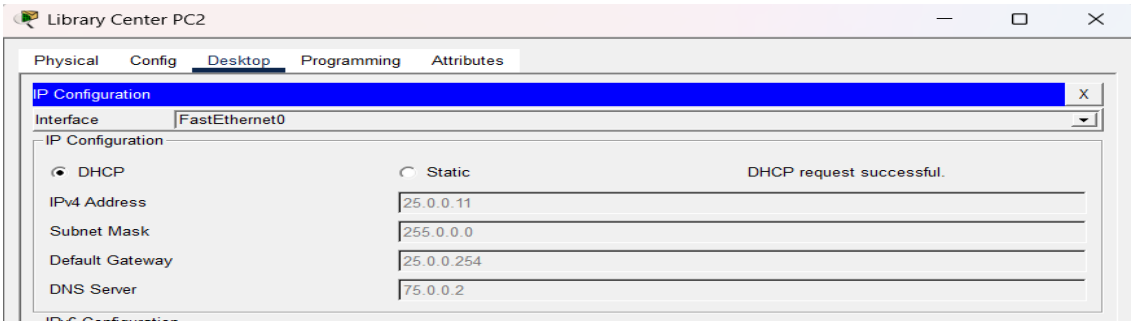



**Figure 10.2: Automatic DHCP IP Assignment to PC2 in Library Center**

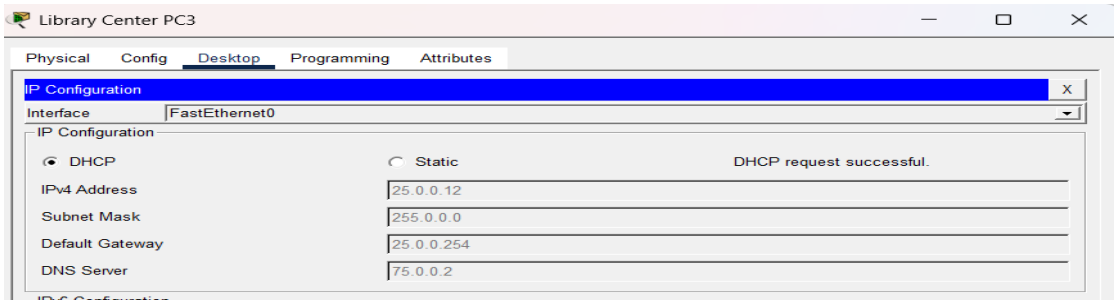



**Figure 10.3: Automatic DHCP IP Assignment to PC3 in Library Center**

## ➢ Faculty Offices DHCP

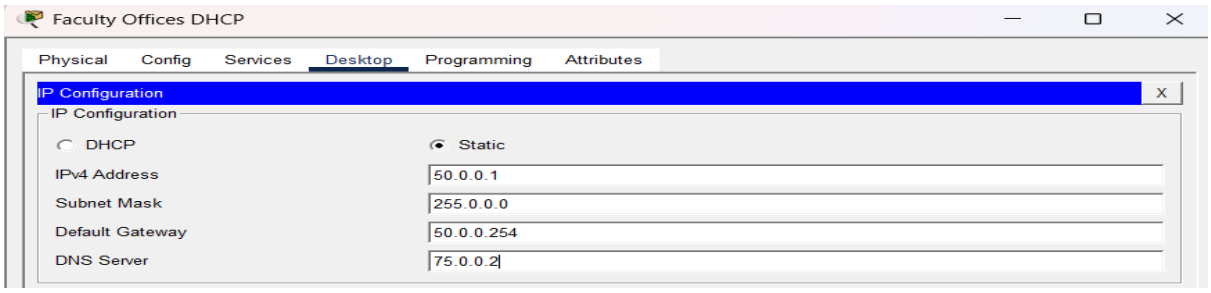



**Figure 11: DHCP Configuration for Faculty Offices**

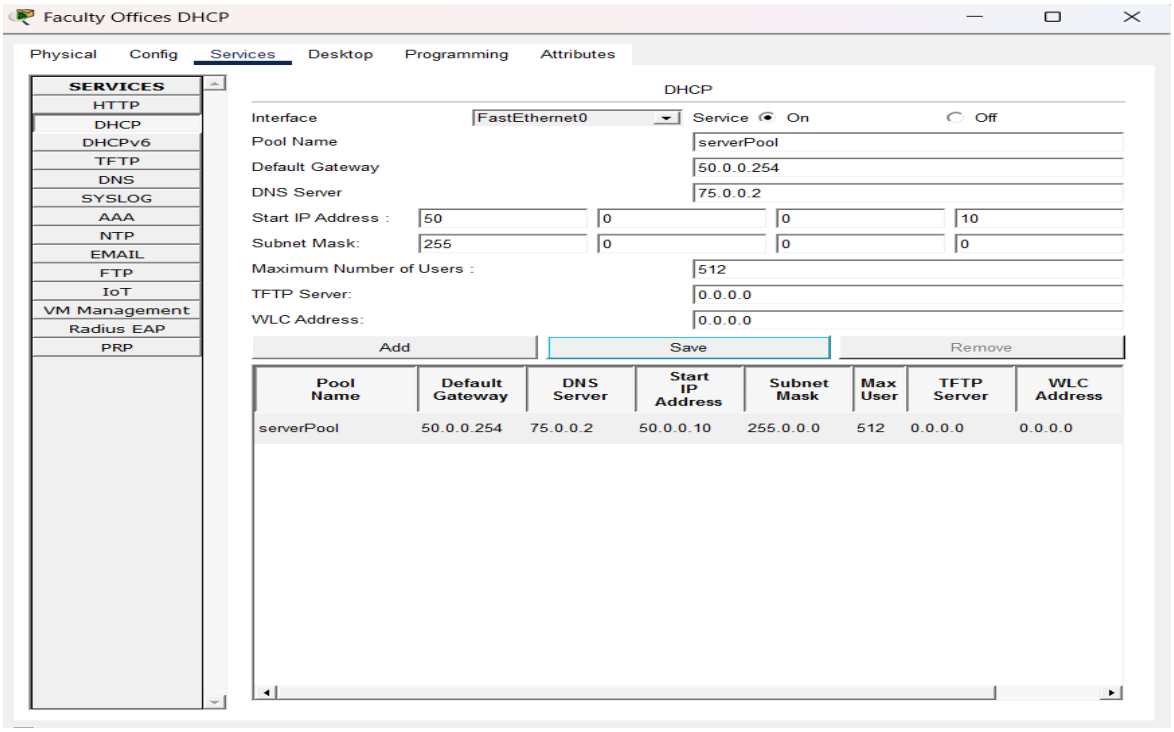



**Figure 12: DHCP Service Running for Faculty Offices**

## ➢ Automatic IP Assignment to PCs in Faculty Offices

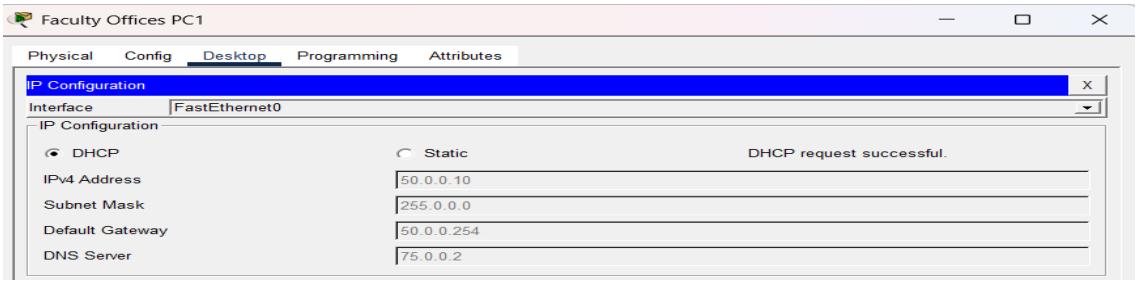



**Figure 12.1: Automatic DHCP IP Assignment to PC1 in Faculty Offices**

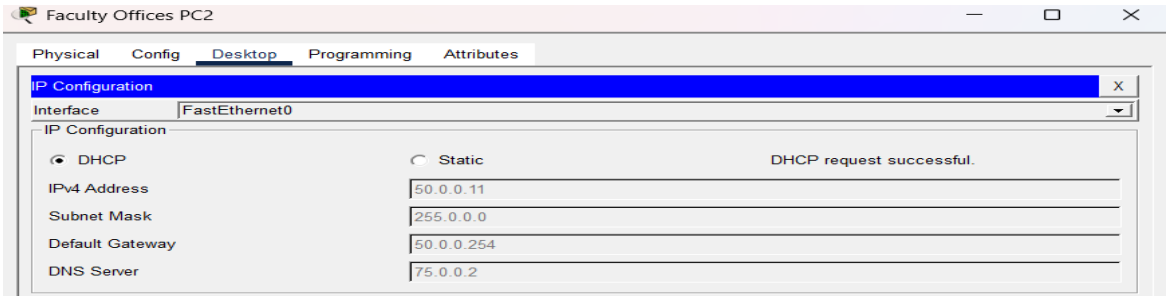



**Figure 12.2: Automatic DHCP IP Assignment to PC2 in Faculty Offices**

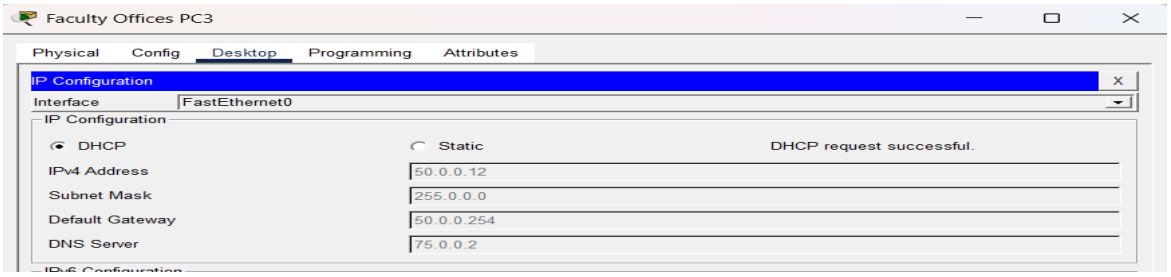



**Figure 12.3: Automatic DHCP IP Assignment to PC3 in Faculty Offices**

## ➢ Main Campus DHCP

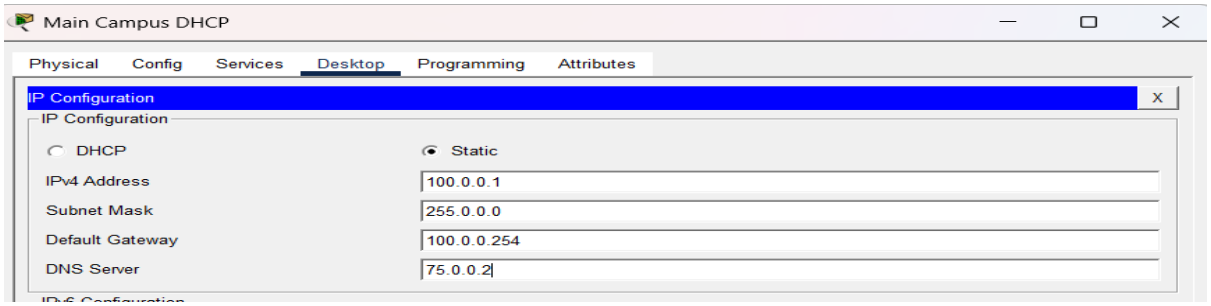



**Figure 13: DHCP Configuration for Main Campus**

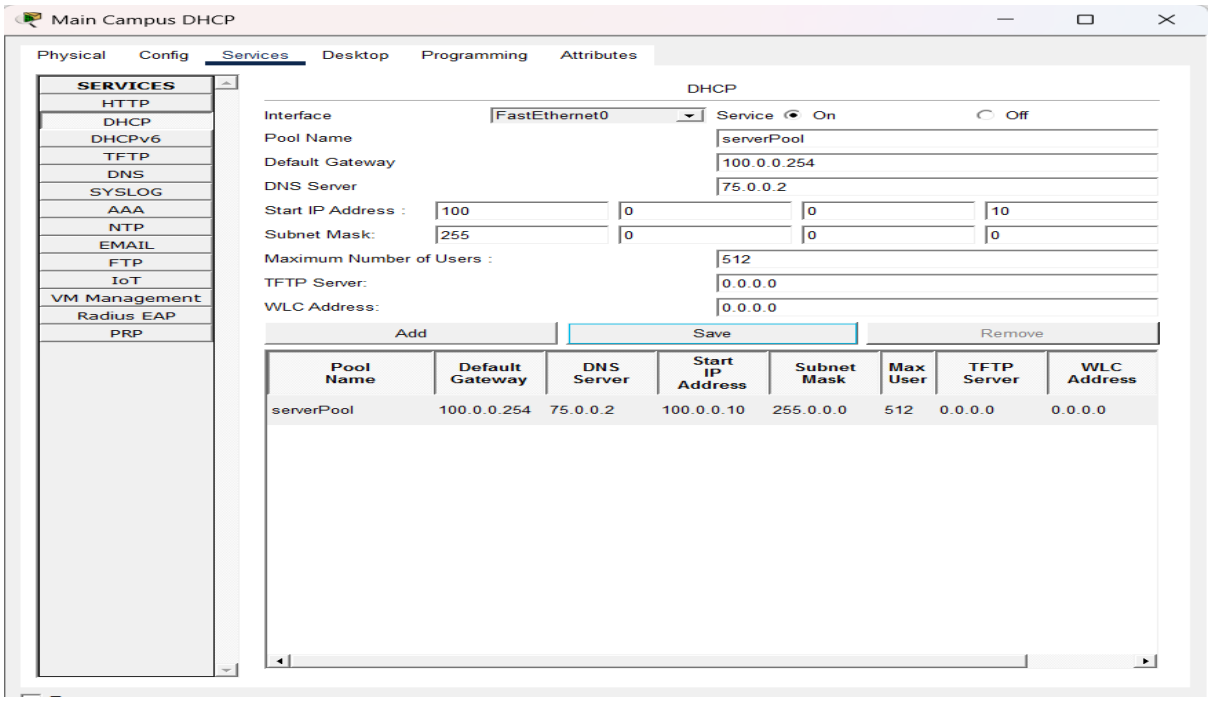



**Figure 14: DHCP Service Running for Main Campus**

## ➢ Automatic IP Assignment to PCs in Main Campus

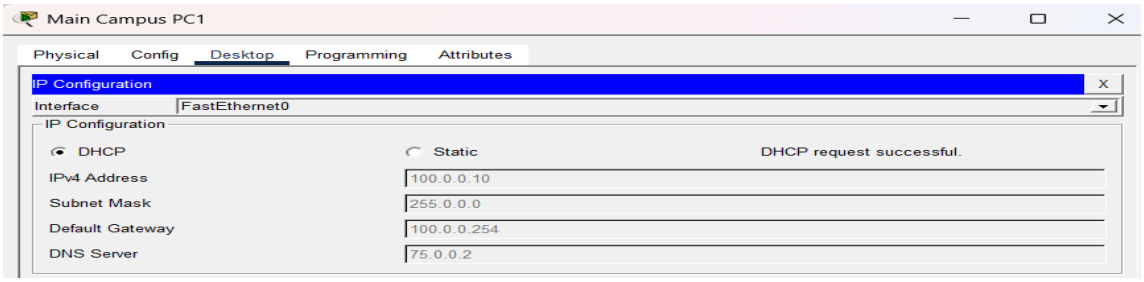



**Figure 14.1: Automatic DHCP IP Assignment to PC1 in Main Campus**

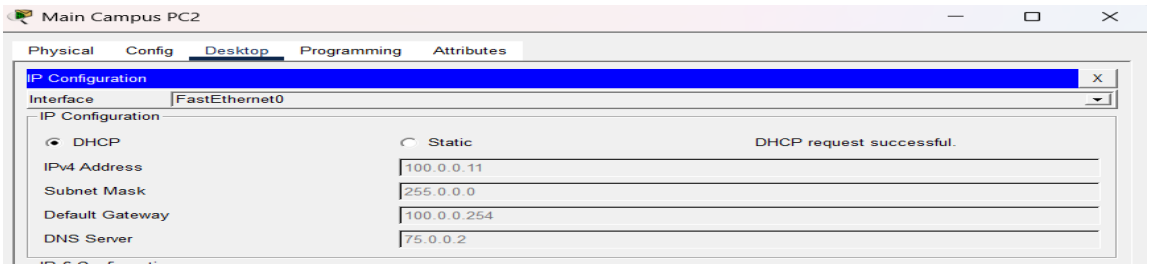



**Figure 14.2: Automatic DHCP IP Assignment to PC2 in Main Campus**

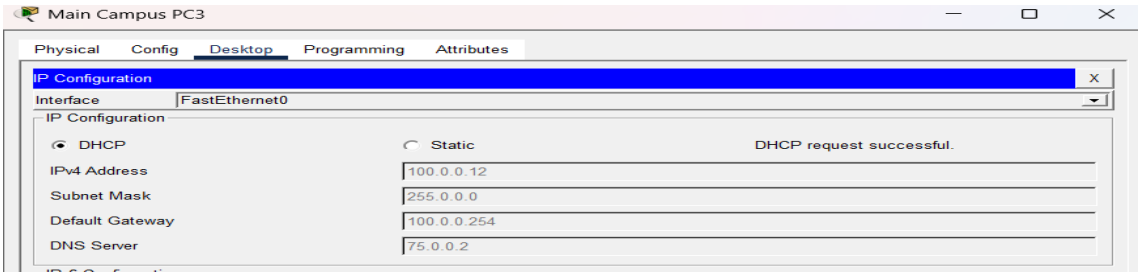



**Figure 14.3: Automatic DHCP IP Assignment to PC3 in Main Campus**

## ➢ Computer Lab DHCP

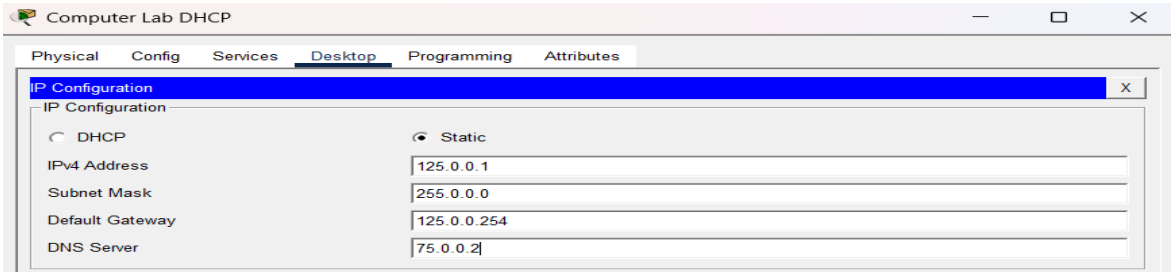



**Figure 15: DHCP Configuration for Computer Lab**

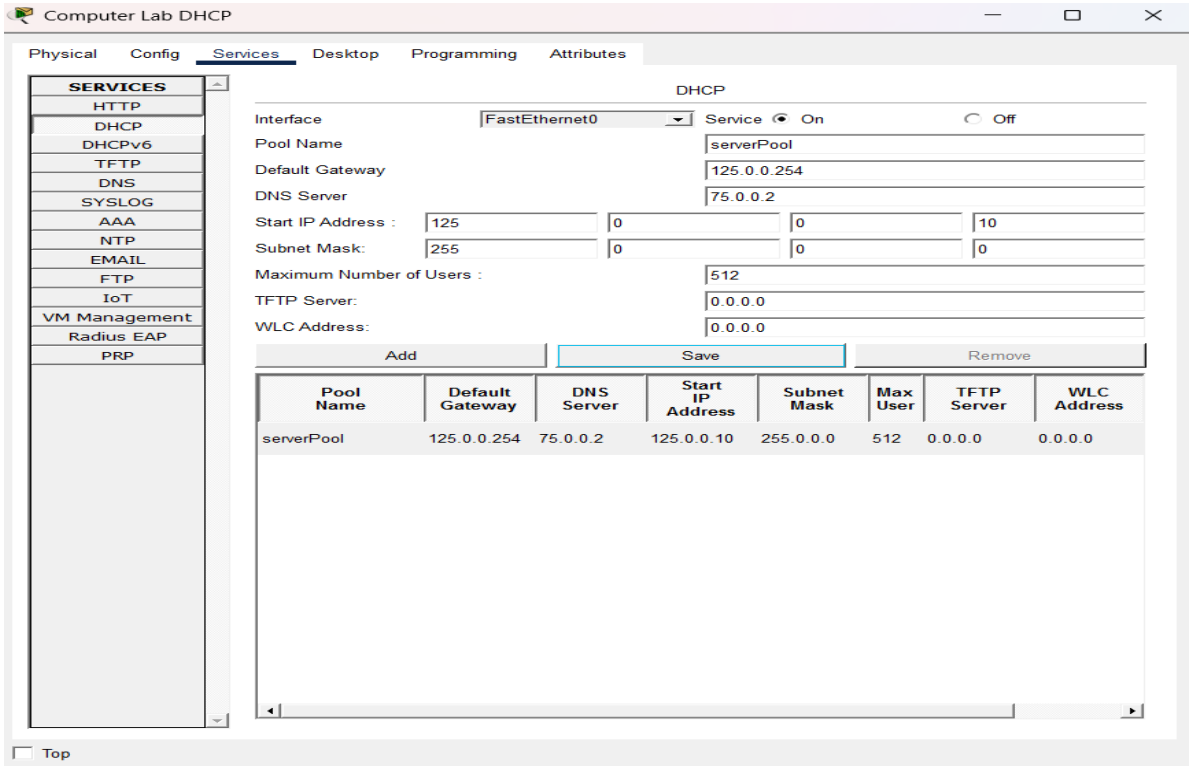



**Figure 16: DHCP Service Running for Computer Lab**

## ➢ Automatic IP Assignment to PCs in Computer Lab

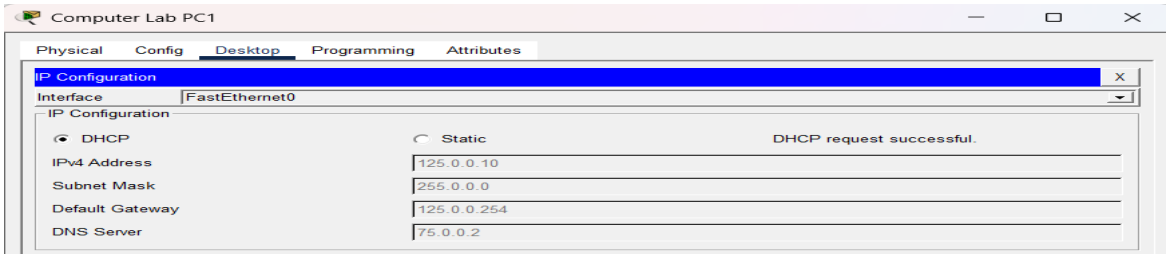



**Figure 16.1: Automatic DHCP IP Assignment to PC1 in Computer Lab**

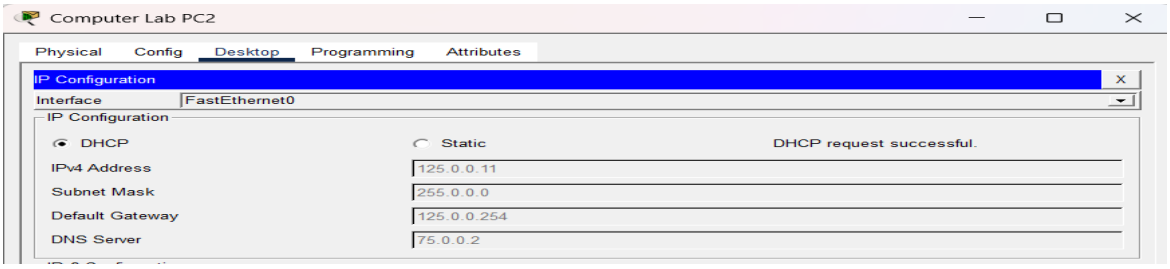



**Figure 16.2: Automatic DHCP IP Assignment to PC2 in Computer Lab**

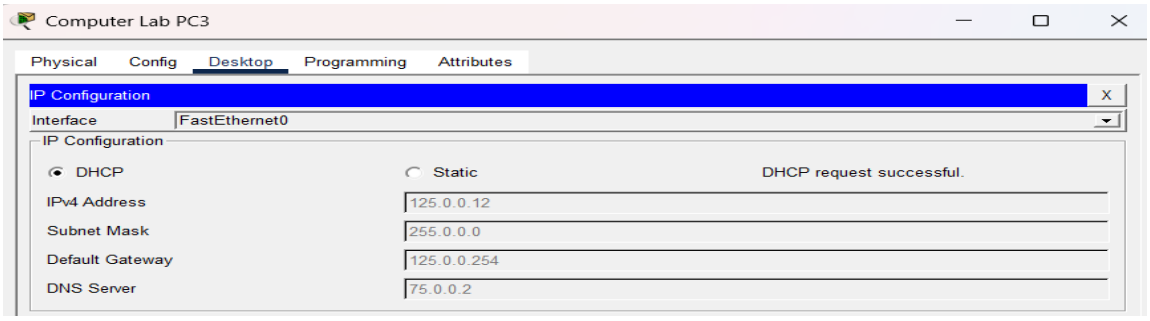



**Figure 16.3: Automatic DHCP IP Assignment to PC3 in Computer Lab**

## 2.5 EIGRP Routing Configuration

Once IP addressing and DHCP configuration were completed, EIGRP (Extended Interior Gateway Routing Protocol) was configured on all routers. EIGRP enabled dynamic routing, allowing routers to automatically exchange routing information and determine optimal paths between different network segments.

This configuration ensured reliable and efficient inter-facility communication across the entire campus network.

## ➢ Eigrp for Administration Block Router

Router>en
Router#config
Router(config)#router eigrp 1
Router(config-router)#network 144.140.0.0 0.0.255.255
Router(config-router)#network 159.150.0.0 0.0.255.255
Router(config-router)#network 174.170.0.0 0.0.255.255
Router(config-router)#network 189.180.0.0 0.0.255.255
Router(config-router)#network 75.0.0.0 0.255.255.255
Router(config-router)#exit

```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 144.140.0.0 0.0.255.255
Router(config-router)#network 159.150.0.0 0.0.255.255
Router(config-router)#network 174.170.0.0 0.0.255.255
Router(config-router)#network 189.180.0.0 0.0.255.255
Router(config-router)#network 75.0.0.0 0.255.255.255
Router(config-router)#exit
Router(config)#
```

Copy Paste

**Figure 17: EIGRP Configuration for Administration Block Router**

## ➢ Eigrp for Library Center Router

Router>en
Router#config
Router(config)#router eigrp 1
Router(config-router)#network 144.140.0.0 0.0.255.255
Router(config-router)#network 25.0.0.0 0.255.255.255
Router(config-router)#exit

```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 144.140.0.0 0.0.255.255
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 144.140.0.1 (Serial2/0) is up: new adjacency

Router(config-router)#network 25.0.0.0 0.255.255.255
Router(config-router)#exit
Router(config)#
```

Copy Paste

**Figure 18: EIGRP Configuration for Library Center Router**

## ➢ Eigrp for Faculty Offices Router

Router>en
Router#config
Router(config)#router eigrp 1
Router(config-router)#network 159.150.0.0 0.0.255.255
Router(config-router)#network 50.0.0.0 0.255.255.255
Router(config-router)#exit

```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 159.150.0.0 0.0.255.255
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 159.150.0.1 (Serial3/0) is up: new adjacency

Router(config-router)#network 50.0.0.0 0.255.255.255
Router(config-router)#exit
Router(config)#
```

Copy Paste

**Figure 19: EIGRP Configuration for Faculty Offices Router**

## ➢ Eigrp for Main Campus Router

Router>en
Router#config
Router(config)#router eigrp 1
Router(config-router)#network 174.170.0.0 0.0.255.255
Router(config-router)#network 100.0.0.0 0.255.255.255
Router(config-router)#exit

```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 174.170.0.0 0.0.255.255
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 174.170.0.1 (Serial2/0) is up: new adjacency

Router(config-router)#network 100.0.0.0 0.255.255.255
Router(config-router)#exit
Router(config)#
```

Copy Paste

**Figure 20: EIGRP Configuration for Main Campus Router**

## ➢ Eigrp for Computer Lab Router

```
Router>en
Router#config
Router(config)#router eigrp 1
Router(config-router)#network 189.180.0.0 0.0.255.255
Router(config-router)#network 125.0.0.0 0.255.255.255
Router(config-router)#
Router(config-router)#exit
```

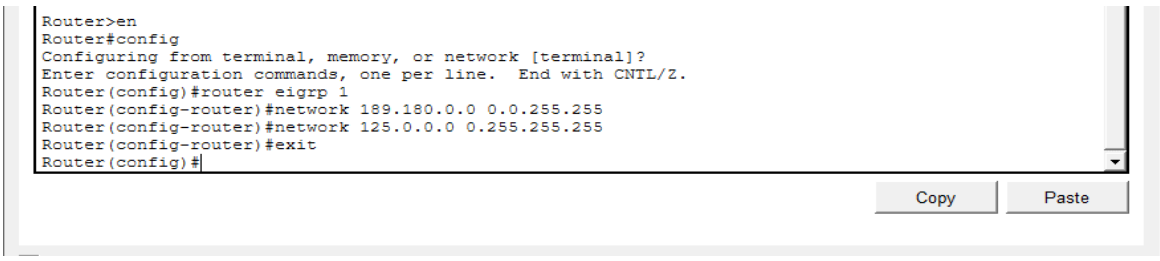



**Figure 21: EIGRP Configuration for Computer Lab Router**

## 2.6 DNS Server Configuration

After completing routing configuration, a centralized DNS server was configured in the Administration Block. The DNS server is responsible for resolving domain names to IP addresses within the campus network.

The domain name **www.universitynetwork.com** was created and mapped to the IP address of the Web Server. Successful DNS resolution ensures that users can access network services using domain names instead of IP addresses.

### ➢ Administration Block DNS

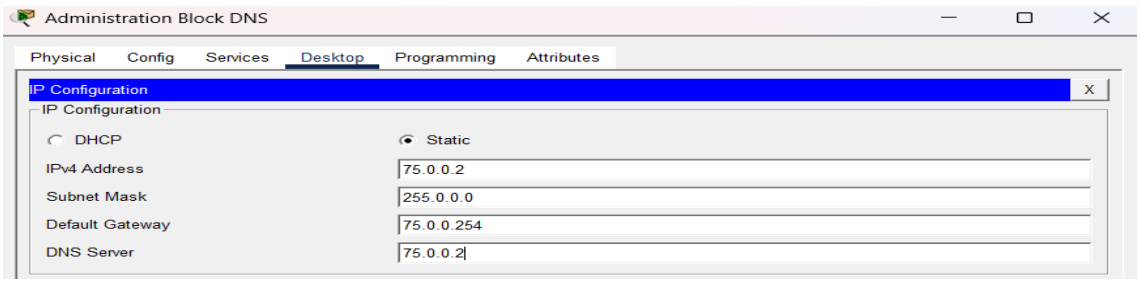



**Figure 22: DNS Server Configuration in Administration Block**

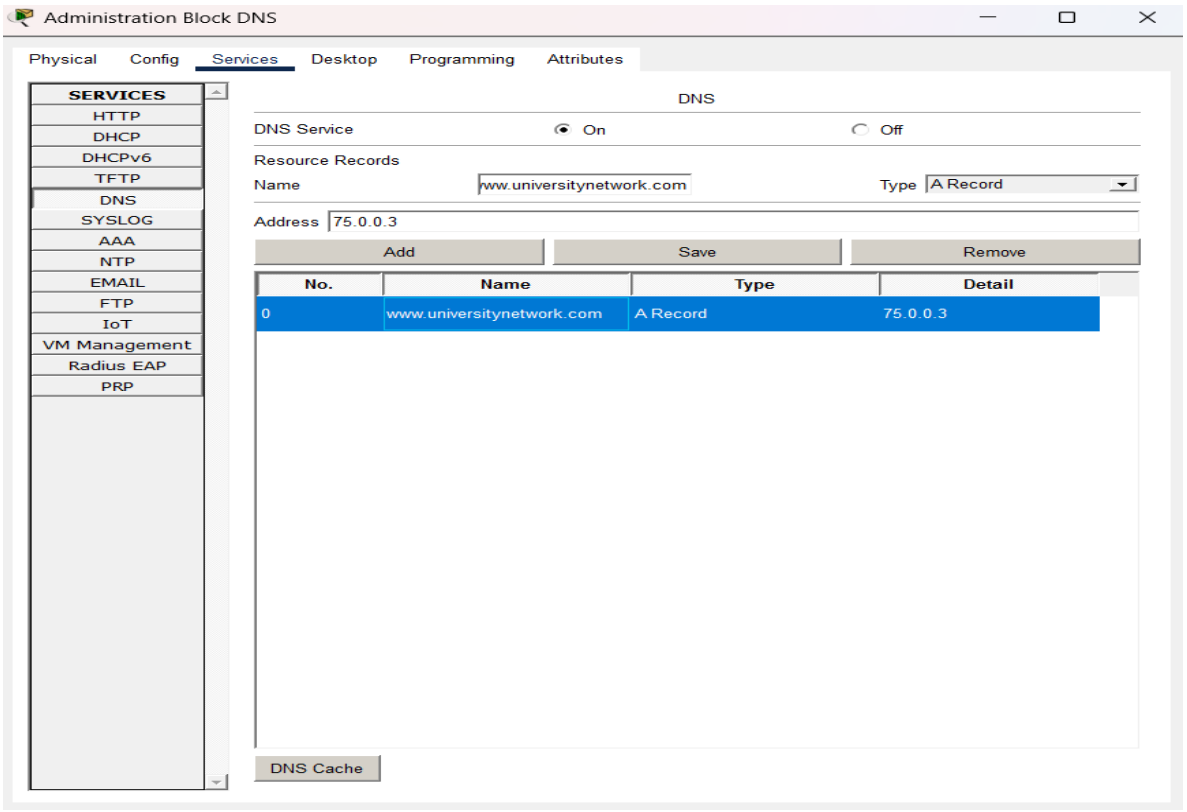



**Figure 23: DHCP Service Running for Administration Block**

## 2.7 Web Server Configuration

Following DNS configuration, a Web Server was configured in the Administration Block to host a webpage accessible across the entire campus network. The web server hosts a webpage displaying the student's full name and student ID as required by the project instructions.

The webpage was successfully accessed from multiple facilities using a standard web browser, confirming proper HTTP service operation and end to end connectivity.

### ➢ Administration Block Web Server

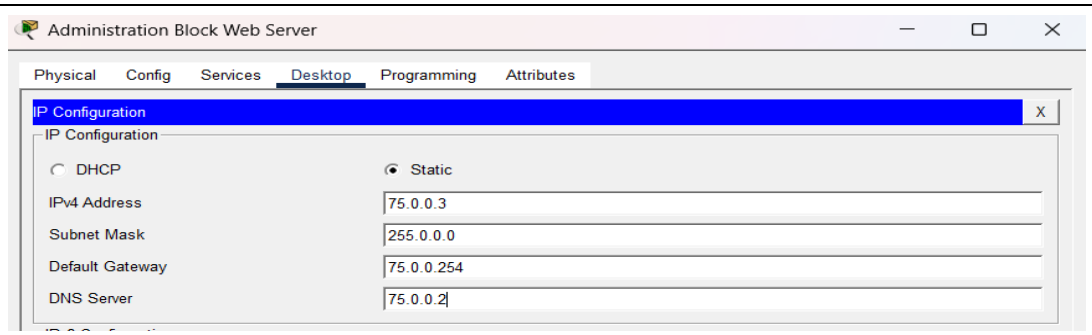


**Figure 22: Web Server Configuration in Administration Block**

# 3. Experimental Results

## 3.1 Inter-Facility Connectivity Test (PC to PC Ping Verification)

After completing router configuration, DHCP assignment, EIGRP routing, DNS and Web server setup, network connectivity was finally verified through practical testing. A ping operation was performed to demonstrate successful inter-facility communication.

- **Ping Command**

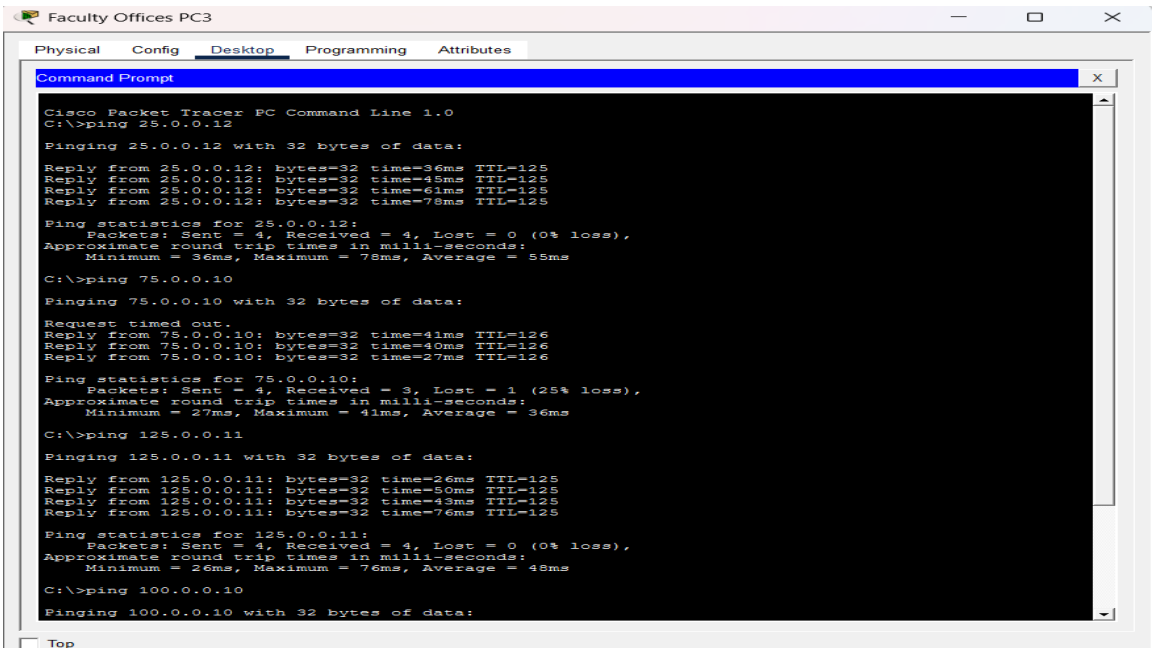



**Figure 23: Ping Command Executed from Faculty Offices PC3**

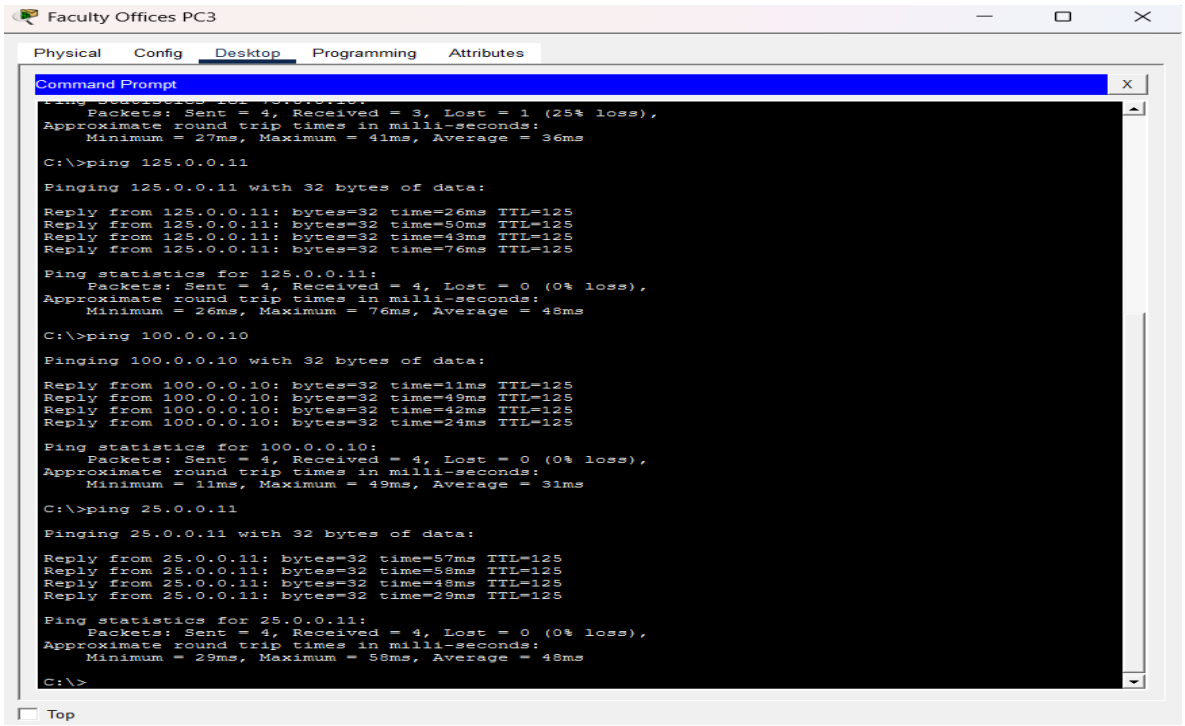



**Figure 24: Ping Command Executed from Faculty Offices PC3**

The test was conducted from the 2nd PC of the Library Center to the 3rd PC of the Administration Block. This test validates that routing, IP addressing, and network services are functioning correctly across different facilities.

The successful ping result confirms proper router configuration, active EIGRP routing, and end-to-end connectivity between the Library Center and Administration BlockA ping operation between the 2nd PC of the Library Center and the 3rd PC of the Administration Block

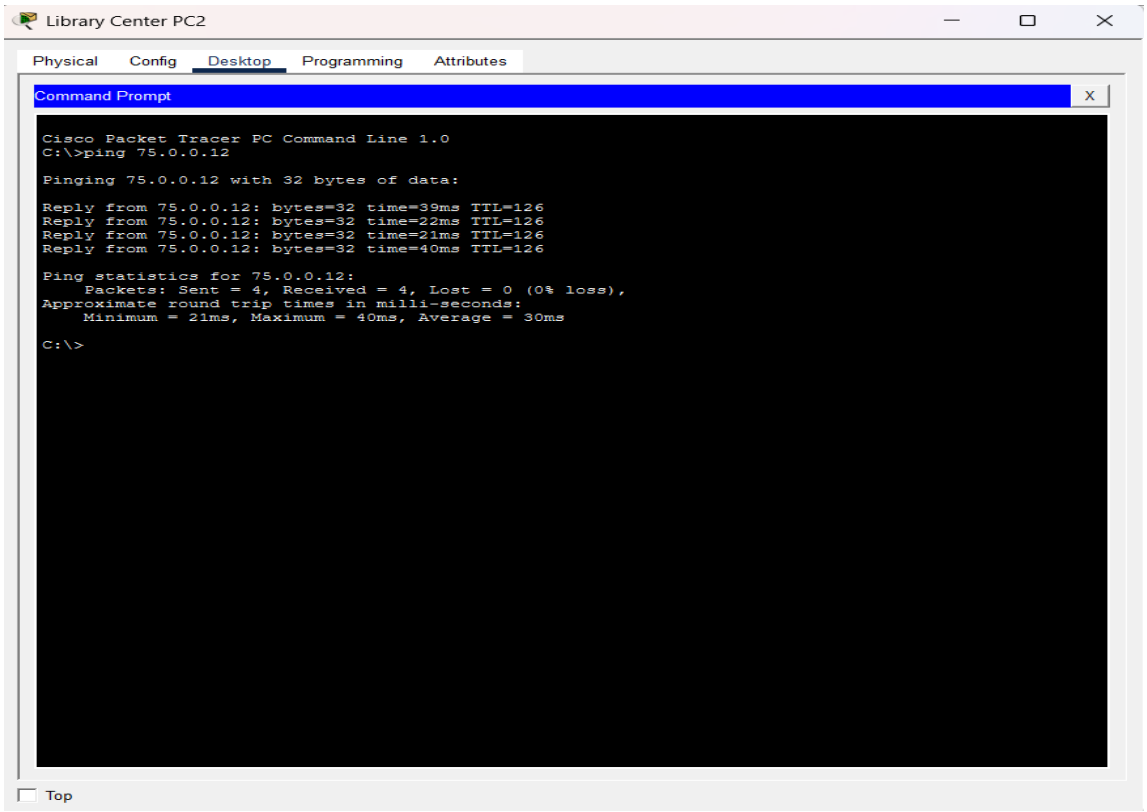



**Figure 25: Ping Command Executed from Library Center PC2**

## ➢ Ping Test Result

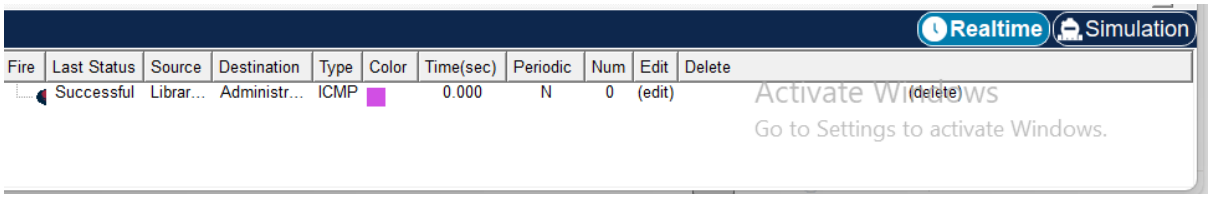



**Figure 26: Successful Ping Result Between Library Center PC2 and Administration Block PC3**

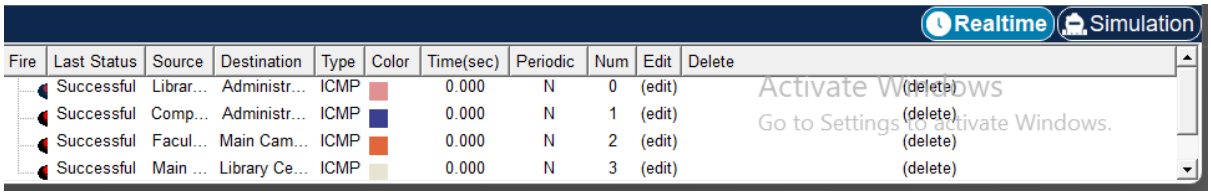



**Figure 27: Successful Ping Result Between Different PCs**

## 3.2 Web Access Verification

The domain **www.universitynetwork.com** was accessed from multiple PCs located in different facilities. The webpage loaded successfully in all cases, verifying proper DNS resolution and web server functionality across the campus network.

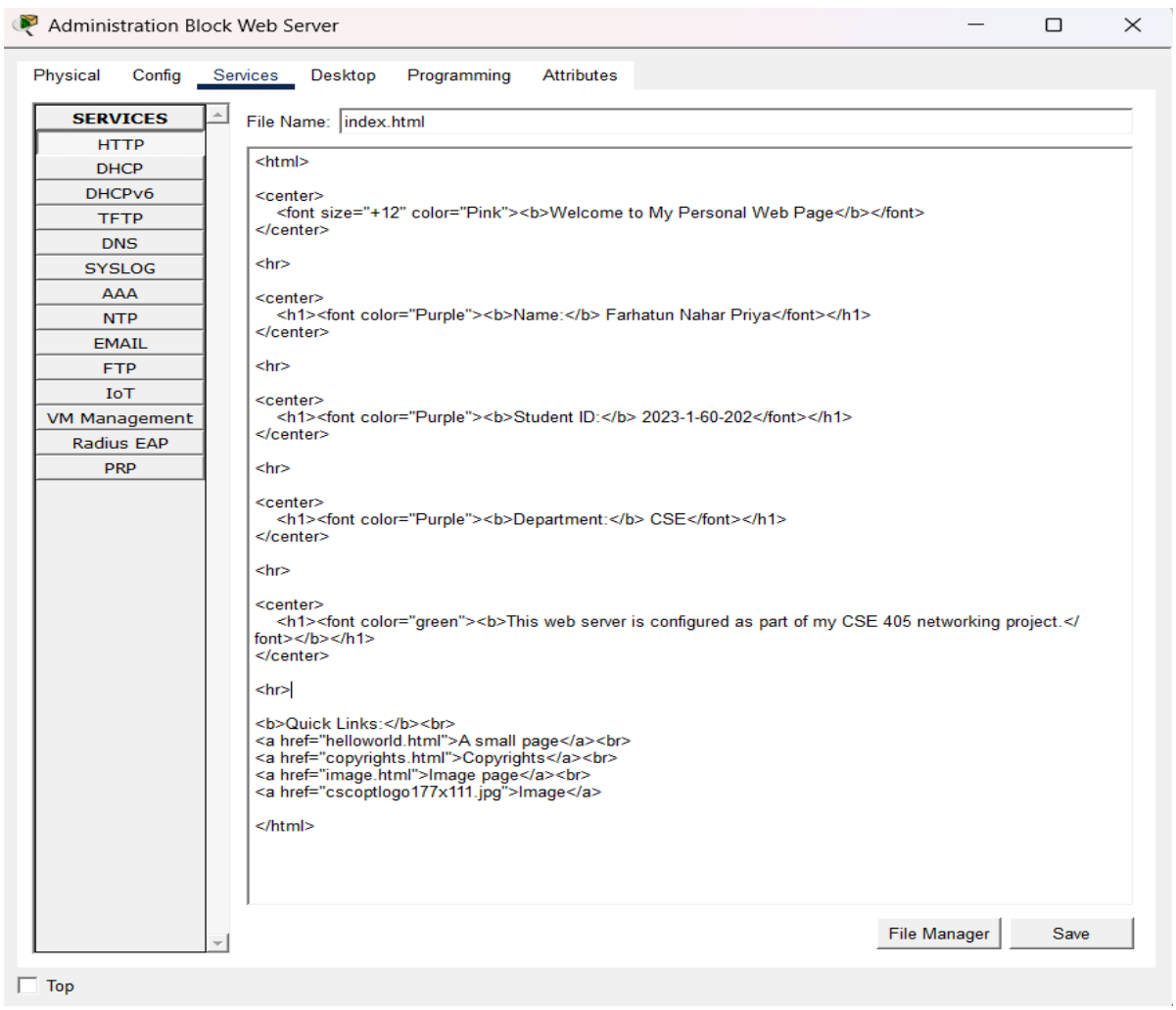



**Figure 28: Administration Block Web Server HTTP Service Configuration**

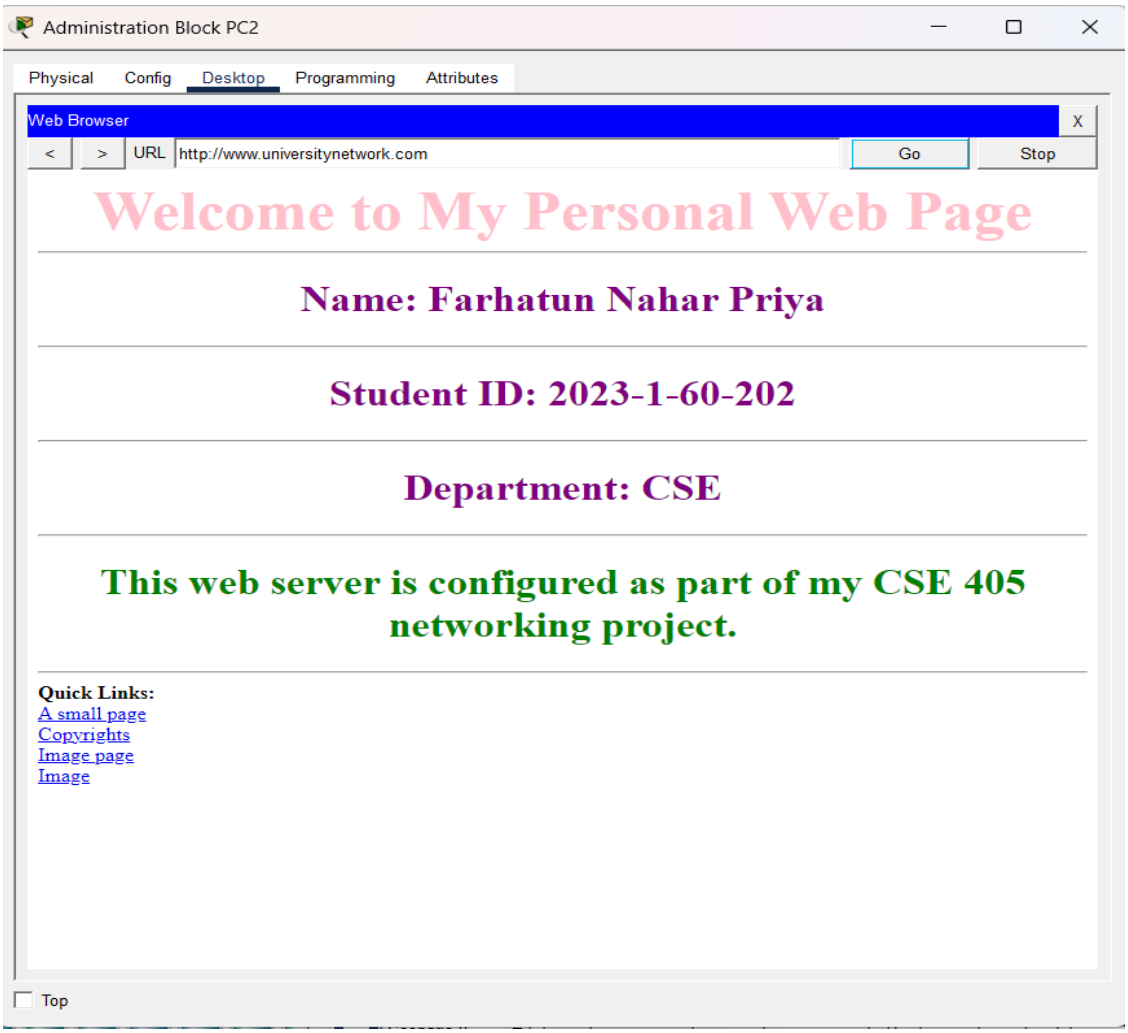



**Figure29: Successful Access of Personal Information Web Page via http://www.universitynetwork.com**

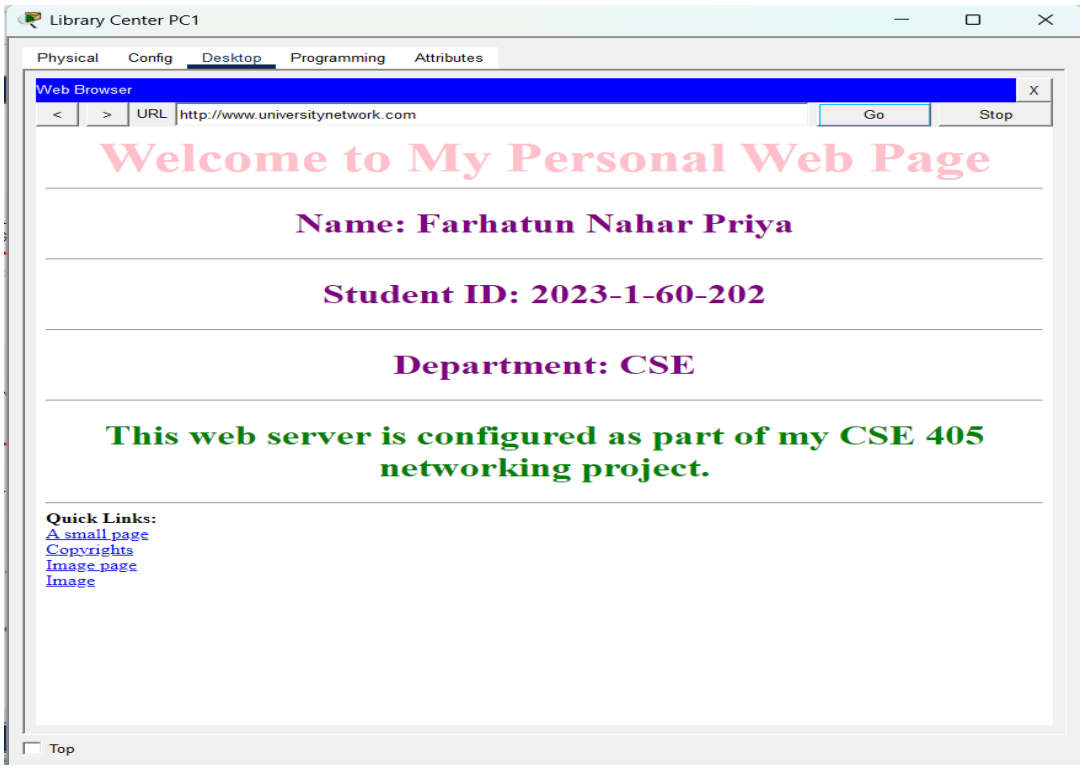



**Figure 29.1: Successful Access of Personal Information Web Page from Library Center PC1 using http://www.universitynetwork.com**

**Figure 29.2: Successful Access of Personal Information Web Page from Faculty Offices PC3 using http://www.universitynetwork.com**

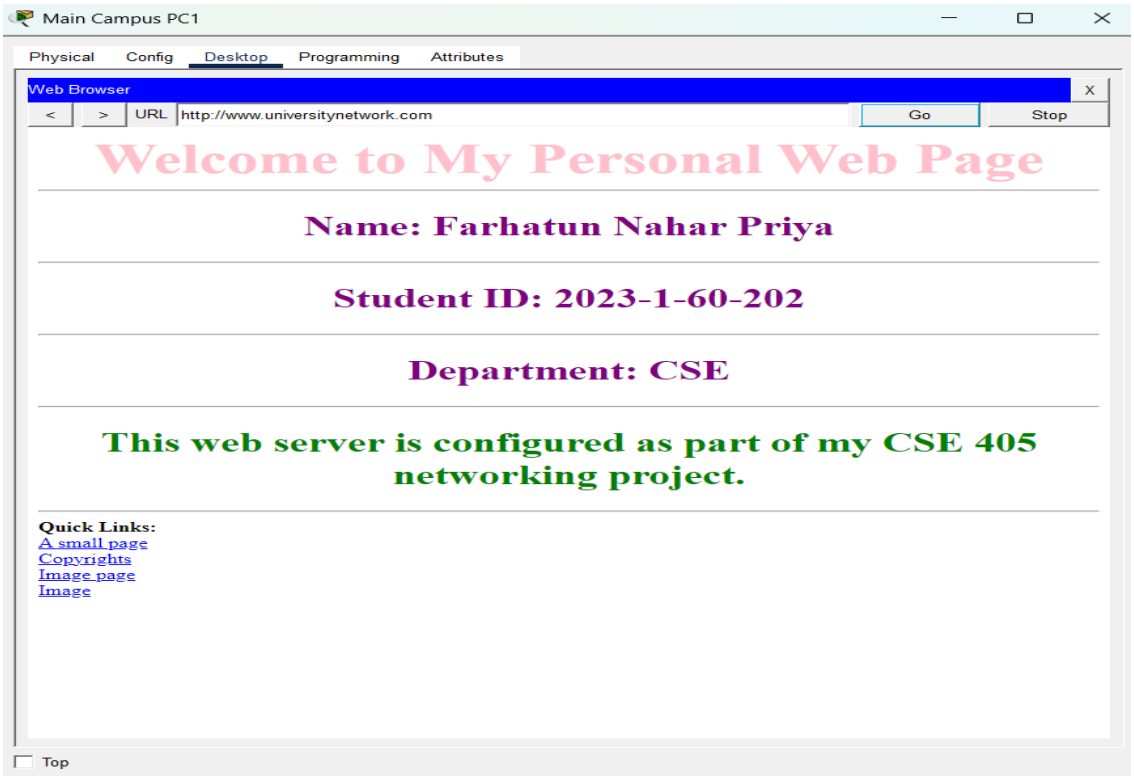



**Figure 29.3: Successful Access of Personal Information Web Page from Main Campus PC1 using http://www.universitynetwork.com**

**Figure 29.4: Successful Access of Personal Information Web Page from Computer Lab PC2 using http://www.universitynetwork.com**

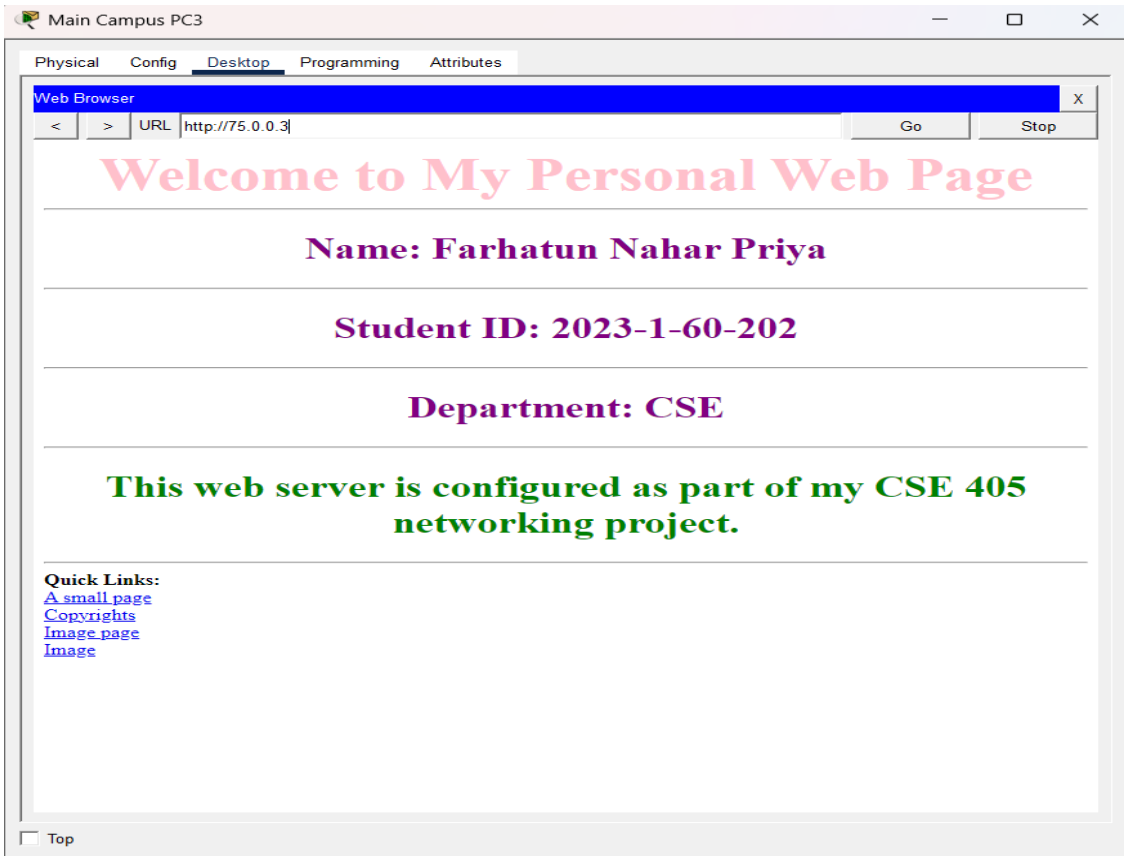



**Figure30: Personal Information Web Page Successfully Accessed from Main Campus PC**

## 4. Conclusion

This project successfully demonstrates the design and implementation of an interconnected campus network using standard networking principles. All project requirements were fulfilled, including dedicated routers for each facility, proper Class A and Class B IP addressing, dynamic routing using EIGRP, automated IP allocation through DHCP, and centralized DNS and Web services.

Experimental testing confirmed reliable inter-facility communication and successful service accessibility. The proposed network design is scalable, efficient and suitable for real world educational environments, effectively applying theoretical knowledge to practical network implementation